Nopal-Sense Project Proposal

Event: IEEE SSCS PICO Open-Source Chipathon 2026
Review: Week 24 Project Proposal Review, June 12, 2026
Track: B - Circuits for Sensors
Secondary relevance: D - AI/LLM-assisted design workflow, for audit and documentation support only
Repository: https://github.com/Jefemaestro33/nopal-sense
Public issue: https://github.com/sscs-ose/sscs-chipathon-2026/issues/16

This document is the reviewable text version of the Nopal-Sense proposal. The slide source in this folder maps the same content to the official four-slide Chipathon proposal template.

1. Team Information

Team name: Nopal-Sense

Role	Name	GitHub	Notes
Design lead	Ernest Zermeño	@Jefemaestro33	Universidad de Guadalajara; individual Track B participant

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2. Project Information

Goal

Nopal-Sense aims to tape out a low-power mixed-signal ASIC for repeatable soil/root-zone impedance spectroscopy using buried electrodes. The research goal is to generate dense, calibrated datasets to test whether root-zone electrical patterns can indicate abnormal biological activity, including fungal or oomycete-related signals after controlled validation.

High-Level Design Proposal

The chip provides programmable AC excitation, electrode-interface readout, filtering, ADC access, and SPI-configurable control. Around the impedance path, the architecture includes autonomous low-power frequency sweeps, synchronized moisture/temperature context, and electrode contact/health checks to flag unreliable soil data caused by poor contact, corrosion, polarization, or cable faults.

V1 is a measurement and data-generation platform, not a pathogen-diagnosis chip. Commercial impedance AFEs such as AD5940/ADuCM355 are useful baselines; Nopal-Sense focuses on the soil-specific embedded layer: buried electrodes, environmental confounders, contact quality, low-power operation, and longitudinal datasets for later biological validation.

Application

Low-cost agricultural root-zone monitoring for irrigation, salinity, and abnormal-condition screening, with a long-term path toward validated fungal/oomycete pattern detection from longitudinal soil impedance datasets.

Long-Term Product Vision

Prior work shows several pieces of this direction are viable: low-cost soil sensor data can classify avocado stress and Phytophthora root rot patterns, Phytophthora zoospores can produce impedance-detectable events in controlled microfluidics, and impedance methods can phenotype roots and disease effects in opaque growth media. Nopal-Sense does not claim those results are already solved in open agricultural soil. Instead, the chip is intended to make repeatable root-zone impedance measurements cheap and standardized enough to collect the datasets needed to test that harder question.

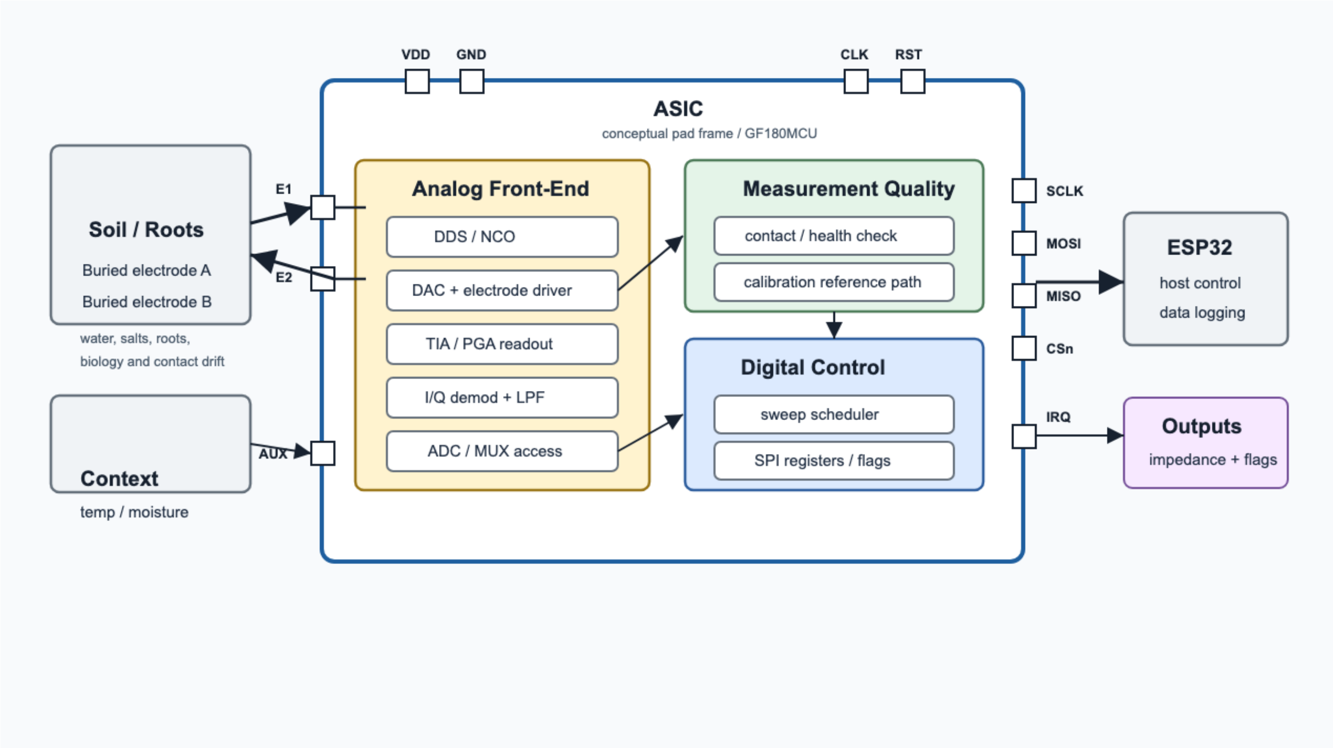

Figure 1: Nopal-Sense ASIC functional architecture

Current Public Progress

- Public GF180MCU project repository created and synchronized with GitHub.
- Frozen public v1 specification, architecture, and 88-pin workshop-slot pin assignment are available in `v1-pico/`.
- 18 functional RTL modules plus top-level integration and workshop wrapper are in the repo.
- Current digital regression reports 185/185 Icarus assertions passing.
- P0 digital integration audit findings have been fixed in RTL.

Remaining Milestones Before Proposal Review

- Publish the proposal deck and link it from the official Chipathon issue.
- Confirm any additional team/collaboration entries.
- Keep the public scope precise: soil/root-zone measurement hub, not plant tissue EIS and not immediate pathogen identification.

Remaining Milestones After Proposal Review

- Add `bridge_controller.v` to decode trigger registers into external SPI, I2C, and 1-Wire transactions.
- Add ADC and sensor behavioral stubs for top-level verification.
- Expand top-level integration coverage beyond the current audited paths.
- Begin analog AFE planning for DDS/DAC, buffer, TIA, I/Q mixer, low-pass filtering, shared SAR ADC, and bandgap/reference.
- Prepare schematic-level review material for the July 3 Schematic Review.

References

- IEEE SSCS Chipathon 2026 repository and participation guidelines: https://github.com/sscs-ose/sscs-chipathon-2026
- Nopal-Sense public repository: https://github.com/Jefemaestro33/nopal-sense
- Nopal-Sense v1 frozen specification: https://github.com/Jefemaestro33/nopal-sense/blob/main/v1-pico/SPEC_FROZEN.md

- Nopal-Sense v1 architecture: https://github.com/Jefemaestro33/nopal-sense/blob/main/v1-pico/ARCHITECTURE.md
- Bukhari et al., "Low-Cost Sensing and Classification for Early Stress and Disease Detection in Avocado Plants", UC Riverside, arXiv:2508.13379, 2025: https://arxiv.org/abs/2508.13379
- Zhang et al., "Microfluidic Biosensors for the Detection of Motile Plant Zoospores", Biosensors, 2025: https://www.mdpi.com/2079-6374/15/3/131
- Corona-Lopez et al., "Electrical impedance tomography as a tool for phenotyping plant roots", Plant Methods, 2019: https://link.springer.com/article/10.1186/s13007-019-0438-4
- Settimi, "Performance of Electrical Spectroscopy using a Resper Probe to Measure the Salinity and Water Content of Concrete or Terrestrial Soil", Annals of Geophysics / arXiv, 2011: https://arxiv.org/abs/1006.4307
- Analog Devices, "AD5940/AD5941 High Precision, Impedance, and Electrochemical Front End", Data Sheet: https://www.analog.com/media/en/technical-documentation/data-sheets/ad5940-5941.pdf
- Analog Devices, "ADuCM355 Precision Analog Microcontroller with Chemical Sensor Interface", Data Sheet: https://www.analog.com/media/en/technical-documentation/data-sheets/aducm355.pdf
- Samouelian et al., "Electrical resistivity survey in soil science: a review", Soil and Tillage Research, 2005: https://doi.org/10.1016/j.still.2004.10.004

3. Team Background

Academic and Technical Experience

- Ernest Zermeño has a biology background and is currently working in bioinformatics research involving CRISPR and experimental biology.
- Technical interests include computational biology, data analysis, embedded systems, and measurement tools for biological and agricultural systems.
- This background motivates the focus on practical soil/root-zone measurements rather than a generic sensor interface.

Work and Application Experience

- Software and applied technology work through zlabstudio.
- Previous work includes exploring FHE-based software development and building practical software systems.
- Focused on turning technical ideas into usable tools, especially where biology, software, data, and measurement systems overlap.

4. Questions, Suggestions, and Doubts

1. What is the recommended must-work scope for v1 if the full architecture is too large for the tapeout schedule?
2. Should v1 reuse an existing open-source/reference SAR ADC instead of a custom ADC design to reduce schedule risk?
3. What electrode protection, ESD strategy, and analog pad usage are recommended for soil probes connected outside the chip package?
4. What minimum AMS validation evidence is expected for the July block-level and top-level simulation reviews?